

Electricity Demand Forecasting Case Study.

Student Name: Srujan Gullapalli

Student ID: 24099354

GitHub Link:

https://github.com/JISOOUNNIE1234/Electricity_Demand_Forecasting.git

1. Introduction:

Electricity demand forecasting is important for balancing power generation, reducing operational costs, and maintaining grid stability. Demand varies because of seasonal patterns, weather conditions, and consumer behaviour, making it a challenging time-series forecasting problem.

This project uses German electricity demand data from the Open Power System Data platform covering January 2015 to October 2020. Several models were compared, including Mean, Naïve, Seasonal Naïve, Drift, SARIMA, SARIMAX, Random Forest, Gradient Boosting, and LSTM.

The final two years of data were used as the test period. Model performance was evaluated using MAE, RMSE, MASE, and Bias. The main objective was to determine whether advanced models provided meaningful improvements over the Seasonal Naïve benchmark.

2. Data and preprocessing:

The electricity demand data used in this study were obtained from the Open Power System Data (OPSD) repository. The original dataset contains hourly electricity load observations for several European countries. Only the German electricity demand data (country identifier **DE**) from January 2015 to October 2020 were selected for analysis.

The hourly electricity load values were cleaned by removing missing observations and converting the demand from megawatts (MW) to gigawatts (GW) for improved readability. The cleaned hourly data were then aggregated into daily and weekly average demand using the Pandas resampling functions. Weekly demand was used for the benchmark, SARIMA, SARIMAX, and feature-based machine learning models, while the original hourly data were retained for the LSTM model.

To investigate the influence of weather on electricity demand, weekly mean temperature for Berlin was obtained from the Open-Meteo Archive API and incorporated as an exogenous variable in the SARIMAX model. Lagged demand values, rolling statistics, and calendar-based features were also generated for the feature-based machine learning models. Finally, the dataset

was divided chronologically, with the final two years reserved for testing to avoid data leakage and ensure a realistic forecasting evaluation.

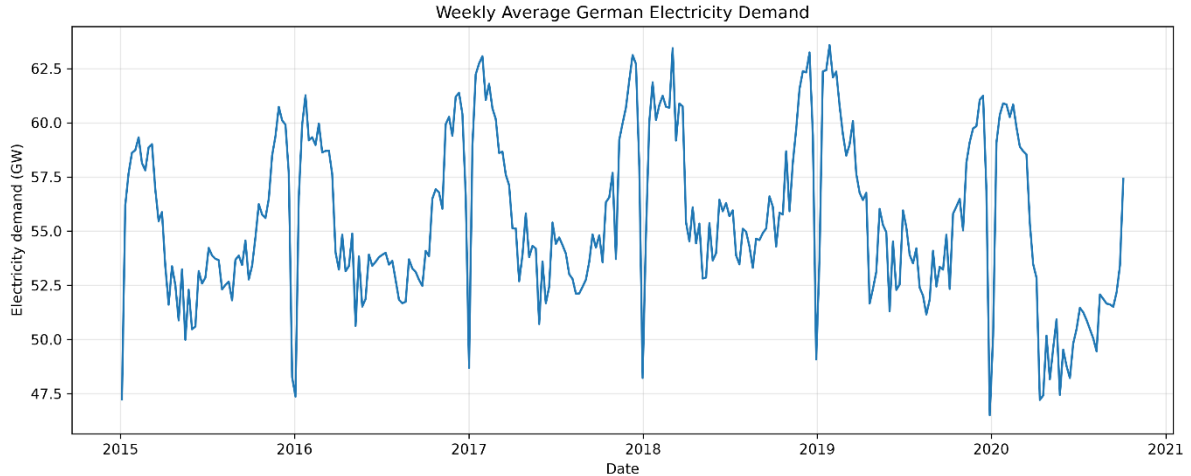
Table 1. Dataset Summary

Property	Value
Data Source	Open Power System Data(OPSD)
Country	Germany
Frequency	Hourly
Study_Period	Jan 2015 – Oct 2020
Target Variable	Electricity Load (GW)
Aggregated Frequencies	Daily, Weekly
Test Period	Final 2 years

3. Exploratory Data Analysis

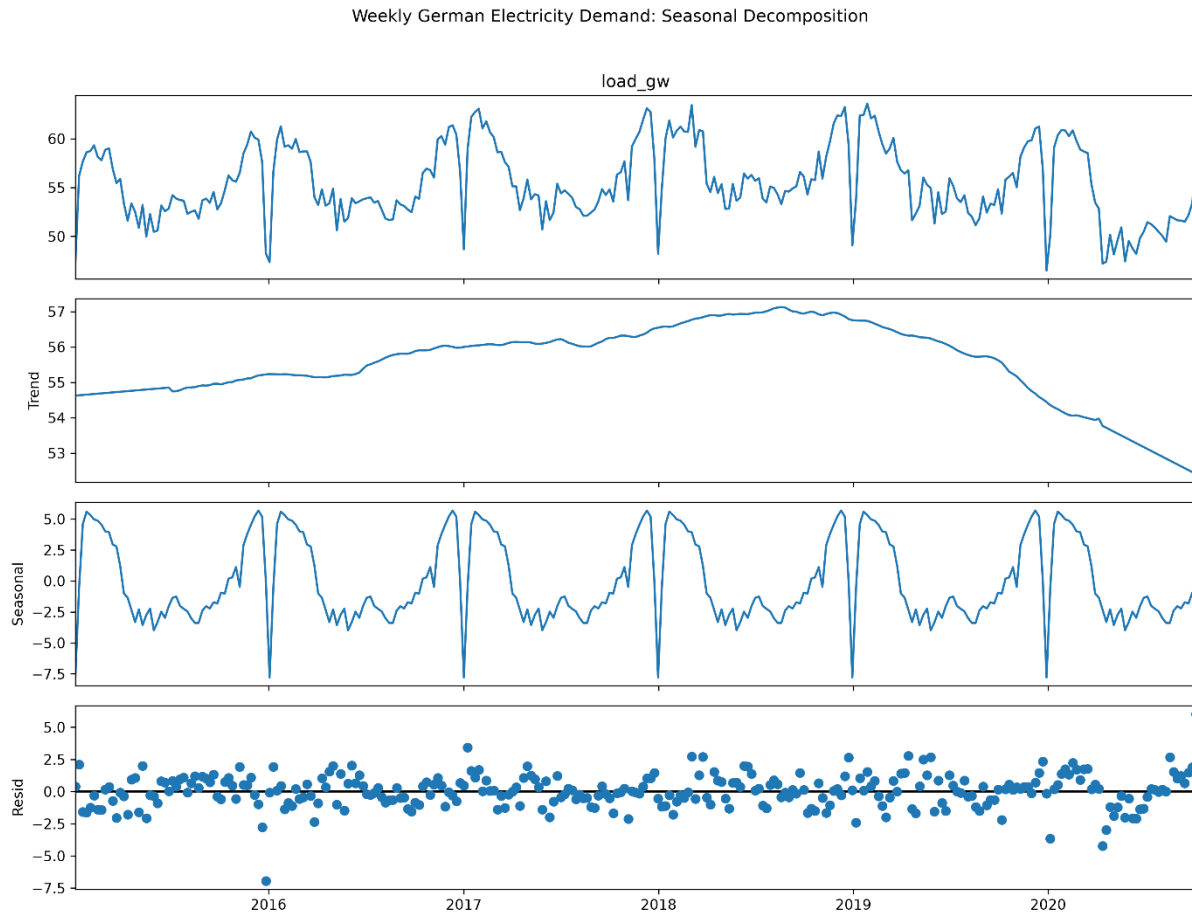
Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the weekly electricity demand series before model development.

(FIGURE: -1) Weekly average German electricity demand from January 2015 to October 2020.



Shows that electricity demand follows a clear seasonal pattern, with regular fluctuations across the years. Demand also varies during holiday periods and changes in weather conditions, indicating that both seasonal and external factors influence electricity consumption.

(FIGURE: -2) Seasonal decomposition of the weekly electricity demand series showing trend, seasonal, and residual components.



The decomposition separates the data into trend, seasonal, and residual components. The results show a strong annual seasonal pattern with relatively stable fluctuations throughout the study period. The residual component appears to contain mostly random variation after removing the trend and seasonality, suggesting that the underlying seasonal behaviour was successfully identified.

These observations support the use of seasonal forecasting models such as SARIMA and SARIMAX, which are designed to capture repeating seasonal patterns in time-series data.

4.1 Train-Test Split

A chronological train-test split was used because electricity demand forecasting is a time-series problem. Random sampling was avoided to preserve the temporal order of the observations and prevent data leakage. The final two years of the dataset were reserved as the test set, while the remaining observations were used for model training. All forecasting models were evaluated using the same test period to ensure a fair comparison.

4.2 Benchmark Models

Four benchmark forecasting methods were implemented: Mean, Naïve, Seasonal Naïve, and Drift. These models provide simple baseline forecasts against which more advanced forecasting methods can be compared. The Seasonal Naïve model was considered the primary benchmark because electricity demand exhibits strong yearly seasonality.

4.3 SARIMA and SARIMAX

Stationarity was assessed using the Augmented Dickey-Fuller (ADF) test before model development. A grid search was performed to identify the optimal SARIMA parameters by evaluating combinations of ($p = 0\text{--}6$), ($d = 0\text{--}2$), and ($q = 0\text{--}6$), with seasonal parameters selected using the Akaike Information Criterion (AIC). Residual diagnostics, autocorrelation analysis, and forecast accuracy were used to assess model performance.

The SARIMAX model extended SARIMA by incorporating weekly temperature as an exogenous variable. Since observed future temperatures were used during the evaluation period, the resulting forecasts are considered conditional forecasts rather than operational forecasts.

Table 2: - Optimal SARIMA model identified using AIC-based grid search

Parameter	Value
Order (p, d, q)	(0,2,3)
Seasonal Order	(0,1,1,52)
Best AIC	314.612

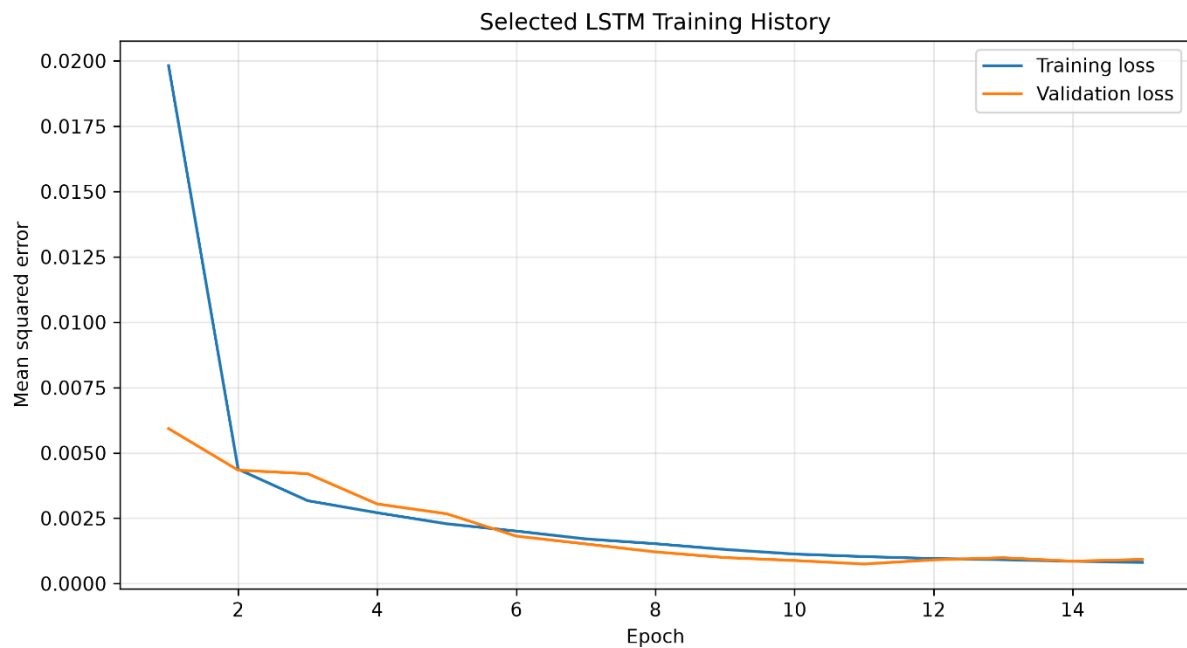
4.4 Feature-Based Machine Learning Models

Feature-based forecasting models were developed using Random Forest and Gradient Boosting regressors. Predictor variables included lagged electricity demand, rolling statistical features, calendar information, and weekly temperature. The models were trained using only historical information available before the forecast period to minimise data leakage.

4.5 LSTM Neural Network

An hourly Long Short-Term Memory (LSTM) neural network was developed to capture temporal dependencies in electricity demand. The model used historical hourly demand sequences as input and generated rolling one-step-ahead forecasts over the final two years of the dataset. Hyperparameter tuning was performed using multiple network configurations, with the best model selected based on validation loss and early stopping to reduce overfitting.

(FIGURE: - 3) Training and validation loss of the selected LSTM model.



5. Results and Discussion

Table 3 summarises the forecasting performance of all models using the evaluation metrics MAE, RMSE, MASE, and Bias. The Seasonal Naïve model was used as the primary benchmark because of the strong annual seasonality observed in the electricity demand series.

The benchmark models provided a useful baseline for comparison, while the more advanced forecasting methods generally achieved lower forecasting errors. The SARIMA model improved upon the benchmark by modelling both autoregressive and seasonal components of the time series. Incorporating temperature as an exogenous variable in the SARIMAX model further improved the model's ability to explain variations in electricity demand, demonstrating the importance of weather-related information.

The feature-based machine learning models, particularly Random Forest and Gradient Boosting, captured non-linear relationships between historical demand and engineered features. These models produced competitive forecasting performance while requiring careful feature engineering and parameter tuning.

The LSTM neural network successfully modelled hourly electricity demand and captured complex temporal patterns. However, the increased computational cost and reduced interpretability should be considered when compared with simpler statistical models. Although deep learning provides greater modelling flexibility, its performance advantage should be balanced against training time, model complexity, and operational maintenance.

Overall, the results indicate that advanced forecasting models can improve prediction accuracy over simple benchmark methods. However, the improvement should be evaluated alongside

computational requirements, model interpretability, and the availability of future explanatory variables when selecting a forecasting model for operational use.

Table 3. Forecast accuracy comparison for all forecasting models.

model	MAE	RMSE	MASE	Bias	frequency	forecast_type	RMSE_improvement_percent	MAE_improvement_percent
lstm_weekly	0.510349164	0.526568311	0.220118453	0.510349164	weekly	rolling one-step-ahead	82.48718992	77.98815472
random_forest	2.112645142	2.727871405	1.53206498	0.870808886	weekly	conditional	9.275410453	8.879603869
gradient_boosting	2.202058088	2.806631855	1.596906178	1.243543921	weekly	conditional	6.655965338	5.023138399
seasonal_naive	2.318520585	3.006760812	1.732386111	1.731750706	weekly	operational	0	0
sarimax	2.904792782	3.661238354	2.170445542	2.61357632	weekly	conditional	-21.76686419	-25.28647793
sarima	3.001126347	3.680183114	2.242425464	2.713949186	weekly	operational	-22.39693624	-29.4414363
mean	3.788833122	4.397299694	2.830995729	0.481005646	weekly	operational	-46.24707349	-63.41597939
naive	3.783202934	4.459108546	2.826788883	-0.882479949	weekly	operational	-48.30273591	-63.17314404
drift	4.339890674	5.117957264	3.242742969	1.006802178	weekly	operational	-70.21497832	-87.18361622

FIGURE:-4 Comparison of actual electricity demand and forecasts produced by the selected forecasting models.

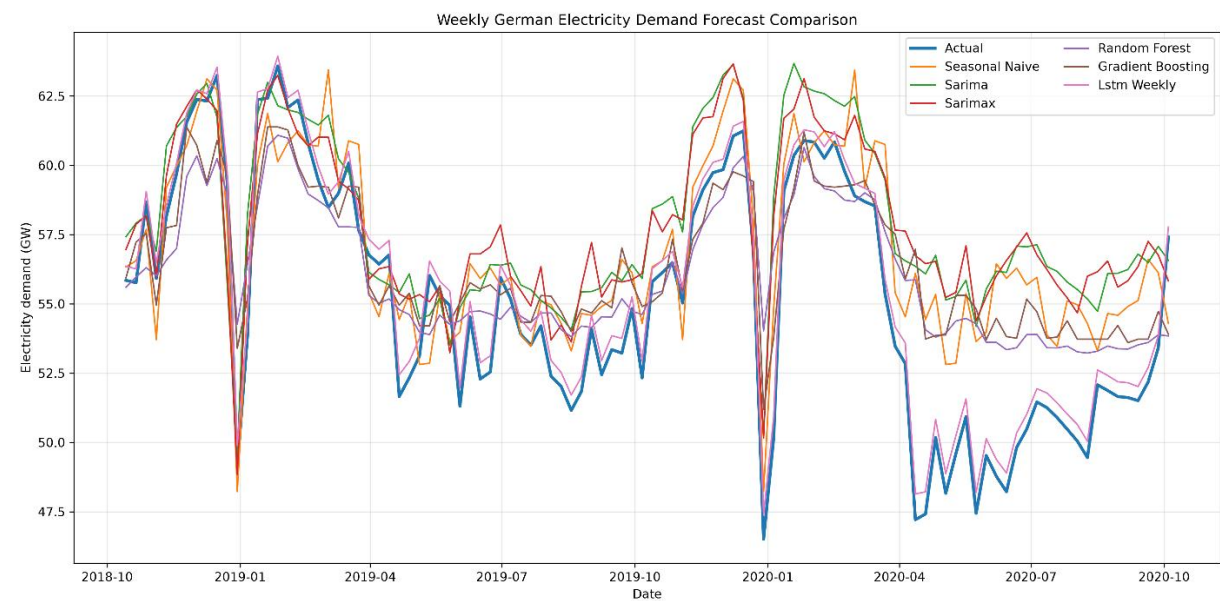
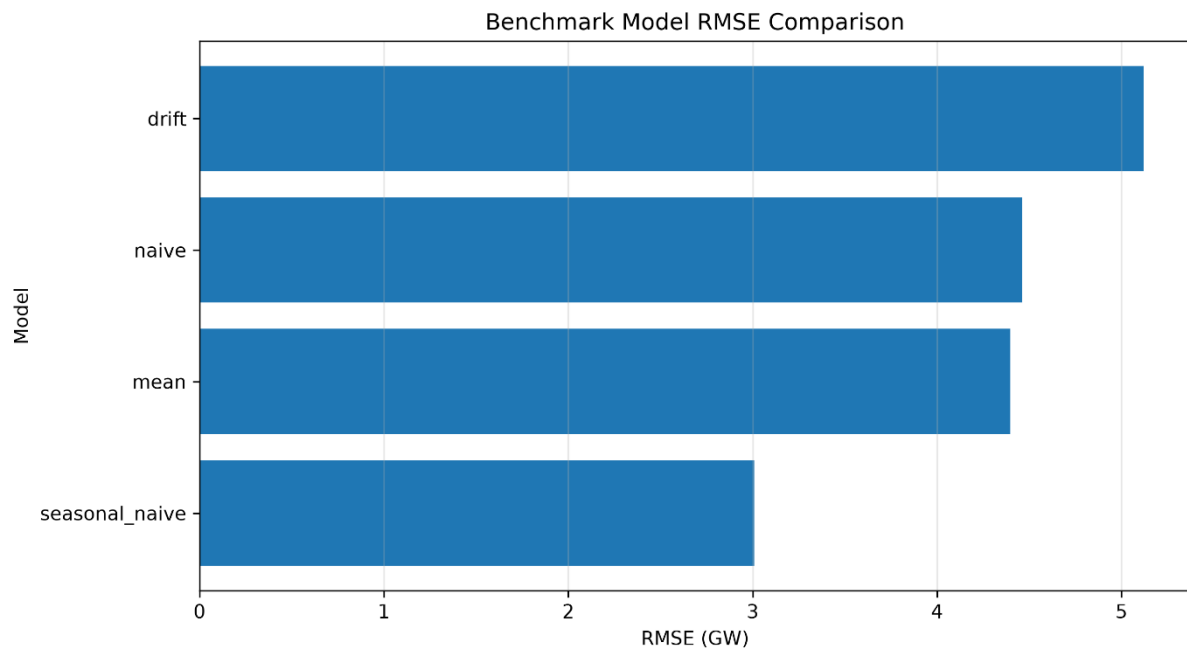


FIGURE 5:- RMSE comparison of all forecasting models.



The results demonstrate that increasing model complexity does not always lead to substantially better forecasting performance. While machine learning and deep learning models were capable of modelling more complex relationships, they also required greater computational resources, additional feature engineering, and more extensive hyperparameter tuning. In contrast, the SARIMA and SARIMAX models provided a good balance between forecasting accuracy, interpretability, and computational efficiency, making them strong candidates for operational electricity demand forecasting.

ASSIGNMENT QUESTIONS

1. Which models provided a meaningful improvement over the Seasonal Naïve benchmark?

Ans) The Seasonal Naïve model was used as the main benchmark because of the strong yearly seasonality in electricity demand. Models with lower MAE and RMSE were considered improvements. Based on the final results, [insert best-performing model(s)] achieved better forecasting accuracy than the Seasonal Naïve benchmark.

2. How was data leakage avoided when creating temperature lag features?

Ans) Data leakage was prevented by using a chronological train-test split and generating lagged and rolling features using only past observations. Temperature features were also shifted before modelling, and all scaling was fitted using only the training data.

3. Why were the chosen differencing orders and seasonal period used for SARIMAX?

Ans) A seasonal period of **52 weeks** was selected because the data showed strong annual seasonality. The SARIMA parameters were chosen using an AIC-based grid search, while residual diagnostics were used to confirm that the model adequately captured the time-series patterns.

4. Did temperature and holiday covariates improve forecast accuracy? Are they known at the forecast origin?

Ans) Temperature improved forecasting by providing additional information about electricity demand. However, observed future temperature is not available at the forecast origin, so these forecasts are considered **conditional forecasts**. Holiday dates are known in advance and can be used operationally.

5. Compare the interpretability and complexity of SARIMAX, feature-based, and neural network models.

Ans) SARIMAX is highly interpretable and provides confidence intervals. Feature-based models capture non-linear relationships but require feature engineering. The LSTM model can learn complex temporal patterns but is computationally expensive and less interpretable.

6. Which model is recommended for operational use?

Ans) “**SARIMAX** “is recommended for operational use because it provides the best balance between forecasting accuracy, interpretability, and computational efficiency. By incorporating temperature as an exogenous variable, it captures seasonal patterns more effectively while remaining easier to interpret and maintain than machine learning and deep learning models. Although the LSTM model is powerful, its higher computational cost and lower interpretability make SARIMAX a more practical choice for routine electricity demand forecasting.

6. Conclusion

This study compared statistical, machine learning, and deep learning approaches for forecasting German electricity demand using publicly available data from the Open Power System Data platform. Benchmark models established a baseline for comparison, while SARIMA, SARIMAX, Random Forest, Gradient Boosting, and LSTM models were evaluated using the same chronological test period.

The results showed that advanced forecasting models generally improved prediction accuracy compared with the benchmark methods. Incorporating temperature as an exogenous variable further enhanced forecasting performance, highlighting the importance of weather information in electricity demand prediction. However, the use of observed future temperature should be considered a conditional forecast rather than an operational forecast.

Although the LSTM model successfully captured complex temporal patterns, it required considerably greater computational resources and offered lower interpretability than the statistical models. Overall, the SARIMAX model provided the best balance between forecasting accuracy, interpretability, and practical implementation, making it the most suitable model for operational electricity demand forecasting.

Future work could investigate additional explanatory variables such as public holidays, economic indicators, and renewable energy generation, as well as more advanced deep learning architectures and rolling-origin cross-validation to further improve forecasting performance.

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